

Team Contributions

Team Member	Contributions
Amanpreet Bhatia	Data preprocessing, EDA, model development, evaluation, report
Mansi Gaur	Data preprocessing, explainability, prediction workflow, PPT

Problem Background

Cardiovascular disease is one of the leading causes of death worldwide, and early detection is critical for improving patient outcomes. Traditional risk assessment methods are often time-consuming and inconsistent across practitioners. Machine learning offers a more consistent, data-driven approach - but for clinical adoption, predictions must be both accurate and interpretable. Clinicians need to understand *why* a patient was flagged, not just *that* they were flagged.

The Problem We Are Solving

This project builds a **Disease Risk Prediction System** that takes patient health data as input and predicts whether a patient is at elevated risk of heart disease. The pipeline is designed to be generalizable to other diseases. Three core challenges are addressed: accurate risk prediction using standard clinical indicators, interpretable predictions suitable for clinical use, and equitable performance across patient subgroups including male and female patients. The system is a **clinical decision support tool**, not a replacement for medical diagnosis.

Methodology

Dataset: The UCI Heart Disease dataset (Janosi et al., 1988) contains 920 patient records with 15 clinical features including age, sex, chest pain type, blood pressure, cholesterol, and maximum heart rate. The multi-class target (0-4) was converted to binary, i.e. 0 for no disease, 1 for disease present.

Preprocessing: Two features with high missingness - ca (~66%) and thal (~53%) were retained due to clinical significance. Missing indicators were added before applying KNN Imputation (k=5). Categorical features were manually encoded using predefined mappings before splitting. All data-driven preprocessing steps like imputation, scaling, and Winsorization, were applied after a stratified 70/15/15 train/validation/test split to prevent data leakage.

Modeling: Four models were trained and evaluated - Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. With the three advanced models tuned using GridSearchCV optimizing for recall, the most critical metric in medical screening. Logistic Regression was selected as the final model for its best overall balanced performance (accuracy 0.87, F1 0.89), highest stability across cross-validation folds (std = 0.021), and direct interpretability through feature coefficients. Tuned ensemble

models achieved marginally higher recall (0.922 vs 0.909) but did not justify the loss of interpretability in a clinical context.

Evaluation & Fairness: On the held-out test set, the final model achieved accuracy 0.87, recall 0.93, F1 0.89, and ROC-AUC 0.92. Subgroup analysis revealed a performance gap - female recall was only 57% at the default threshold of 0.50 versus 91% for males. Threshold optimization lowered the decision boundary to 0.45, improving female recall to 100% and overall recall to 0.93 with no increase in false alarms.

Explainability: SHAP analysis identified chest pain type, ST depression (oldpeak), and exercise-induced angina as the three strongest predictors. These were consistent across EDA, model coefficients, and SHAP values. Individual patient predictions are explained through SHAP waterfall plots showing each feature's contribution.

Prediction Workflow: An end-to-end pipeline accepts raw patient data, applies all preprocessing steps automatically, and returns a risk prediction with SHAP-based explanation. Testing confirmed clinically sensible results - 95.8% disease probability for a high-risk profile and 1.6% for a low-risk profile.

References

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